Seldon Manifolds — Theory

The Geometry of Reachable Futures

Matthew Parslow
Independent Researcher

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Abstract

Graphic Equalisation requires only admissible continuation and deterministic collapse. It does not require a graph or containing frame to represent unrealised futures as an object. This paper studies the additional structure that appears when a frame represents the futures that remain distinguishably reachable under counterfactual admissible interventions and represented future boundary conditions. The resulting object is the *Seldon manifold*: a frame-relative, horizon-relative, resolution-relative representation of reachable future state or history.

The construction is deterministic at substrate level. A Seldon manifold is not a probability distribution over multiple outcomes of one causally complete antecedent. It is a representation of different deterministic continuations obtained under different admissible interventions, future interactions, or unresolved containing conditions represented by the frame. If all causally relevant future conditions and interventions are fixed, the represented continuation collapses to one trajectory.

The manifold need not be smooth or continuous; "manifold" denotes structured reachable geometry. Its dimensions are instantiated only where represented, its values are discrete at operative resolution, and its quantisation may change dynamically. Two distinctions are developed at length: what such a projection is a projection *over* — the *undetermined*, which collapse has yet to fix, as against the *undisclosed*, which is realised but not held by this frame — and *causal depth*, the additional determining structure supplied by containing frames rather than by earlier times. Goals, where represented, define acceptable subsets; Muchness measures retained distinguishable future possibility; irreversibility removes reachable distinctions; and control deforms the reachable geometry. No primitive is added to graphhood.

1 Scope and relationship to prior work

Reachable sets, viability kernels, capture basins, invariant sets, dynamic programming, Hamilton–Jacobi reachability, and model-predictive control are established parts of control theory and applied mathematics (Bellman, 1957; Aubin, 1990, 1991; Mitchell et al., 2005; Bansal et al., 2017; Rawlings et al., 2017); this paper claims none of them as new, and its contribution is their placement inside the Graphic Equalisation frame architecture, stated in full in Section 21.

2 Dependency on Graphic Equalisation

Reading for the claim. As in the foundational paper, the components used here are individually established and cited as such; the claim is their arrangement, which is not reachable by evaluating those components separately, since they are stipulated to be known. The composition's own falsifiability commitment is stated in the foundational paper and is inherited here.

Let

$$G_t = (g_t, r_t, \Gamma_t)$$

be a realised graph in individuating frame W_k, and write

$$\mathcal{F}_{k,t} = (G_t, W_k).$$

Graphic Equalisation supplies a deterministic realised transition once the causally complete antecedent configuration is fixed. It also distinguishes immediate admissible possibility from realised collapse:

$$G_t \longrightarrow \mathcal{A}(\mathcal{F}_t) \xrightarrow{\mathcal{C}_t} G_{t+1}.$$

This paper asks what follows when a frame represents recursively reachable continuations over a horizon rather than only the realised next step.

3 Determinism and counterfactual reachability

Proposition 3.1 (Deterministic reachability)**.** *A Seldon manifold does not require stochastic substrate dynamics. For a fixed causally complete antecedent configuration and a fixed future sequence of admissible interventions and containing interactions, Graphic Equalisation yields one realised continuation. Multiple points or histories in a Seldon manifold correspond to different counterfactual future conditions represented by the frame, not multiple random outcomes of one identical complete cause.*

Let $\mathbf{u} = (u_t, \ldots, u_{t+h-1})$ denote an admissible intervention sequence and $\mathbf{b} = (b_t, \ldots, b_{t+h-1})$ a represented sequence of future boundary conditions or containing interactions. For a deterministic transition law T, a future history is generated recursively by

$$G_{j+1} = T(G_j, u_j, b_j).$$

The frame need not know which $\mathbf{b}$ will be realised. Its uncertainty about that sequence may be represented probabilistically if useful, but probability is an epistemic overlay on the reachable family rather than a primitive of the Graph Law.

Corollary 3.2 (Singleton under complete fixing)**.** *If the frame fixes the causally complete antecedent, all future interventions, and all causally operative future containing conditions over horizon h, then the Seldon manifold contains one distinguishable continuation at that resolution.*

This distinction is important because classical reachability and viability theory often permit disturbances, controls, or differential inclusions to generate sets of possible states (Aubin, 1991; Mitchell et al., 2005). The Seldon construction can represent the same set-valued geometry without interpreting that set as fundamental stochasticity.

4 Standing of the construction

Before the definitions, a boundary on what is being claimed, because the natural reading of what follows is stronger than the position of this paper.

Remark (Succession does not require an observer). Realised change, and the succession its enabling-order carries, exist independently of any frame representing them. Graphic Equalisation withholds even a ratchet-mechanism over the changes — succession is a relation carried by which change enables which, not an engine the substrate runs — but that order is substrate-level regardless of any observer. Seldon manifolds, by contrast, are *optional observer or agent machinery* for representing reachable unresolved continuations. Nature does not have to run a Seldon manifold in order to evolve.

Remark (The horizon is doubly observer-relative). The horizon h is not a substrate quantity. By Graphic Equalisation's Axiom on time, the substrate has succession and no time dimension; a temporal dimension exists only where a frame instantiates one to represent that succession. So h is measured in a dimension the observing frame constructed, and indexes a manifold the same frame constructed. Both the ruler and the thing measured are observer machinery, and neither is a claim about what the world contains.

Remark (Higher machinery is frame-relative, not substrate-required). The same holds of every construct in this paper and its successors: manifolds, undetermined counts, inferred futures, Narrative, and Narrativium are first-class quantities *only in frames that instantiate the relations defining them.* They are not universal substrate requirements, and their absence from a frame is not a deficiency of that frame's dynamics.

These remarks are load-bearing for the falsifiability commitments in Section 22: a claim about what an agent's representation supports is not a claim about what the world must contain.

5 Dimensions, values, and resolution

Definition 5.1 (Represented dimension)**.** *A dimension D of a Seldon representation exists in frame $\mathcal{F}$ only where the frame instantiates the relation needed to distinguish values along D. If that relation is absent, D is undefined in that frame; it is not present with value zero.*

Thus

$$D : \text{NULL} \longrightarrow D \in \mathcal{F}$$

is a dimensional instantiation, whereas

$$D = 0$$

is a value statement made only after D exists. This preserves the dimension/value distinction established by Graphic Equalisation.

Definition 5.2 (Resolution)**.** *Resolution ρ specifies which distinctions on instantiated dimensions the frame currently treats as distinguishable in represented future structure.*

Resolution is discrete but may vary dynamically:

$$\rho = \rho(x, h, t, D).$$

No globally fixed lattice, common scale, or continuous coordinate system is assumed. A grounding may map integer state indices into arbitrary physical scales.

Let $\sim_\rho$ denote "indistinguishable at resolution ρ". The quotient

$$\mathcal{Q}_\rho(X) = X/\!\sim_\rho$$

collects states or histories into the distinctions the frame can currently resolve.

6 The Seldon manifold

Definition 6.1 (Seldon manifold)**.** *For frame $\mathcal{F}_t$, horizon h, and resolution ρ, the Seldon manifold*

$$\boxed{\mathcal{S}_{h,\rho}(\mathcal{F}_t)}$$

is the structured representation, at resolution ρ, of future states or histories reachable from $\mathcal{F}_t$ over horizon h under the admissible counterfactual intervention sequences and represented future containing conditions carried by that frame.

Equivalently, if $\mathcal{U}_{t:h}$ is the represented family of admissible intervention sequences and $\mathcal{B}_{t:h}$ the represented family of future boundary-condition sequences, then schematically

$$\boxed{\mathcal{S}_{h,\rho}(\mathcal{F}_t) = \mathcal{Q}_\rho(\{\mathcal{H}_{t:t+h}(\mathbf{u},\mathbf{b}) : \mathbf{u} \in \mathcal{U}_{t:h},\ \mathbf{b} \in \mathcal{B}_{t:h}\})\,.}$$

The term *manifold* is operational rather than differential-geometric. Depending on the grounding, $\mathcal{S}$ may be represented as a reachable set, graph, viability kernel, capture basin, branching history structure, automaton, cell complex, or another object preserving the relevant reachability relations. Viability kernels and reachable sets are therefore direct mathematical predecessors (Aubin, 1990, 1991; Szolnoki, 2000; Mitchell et al., 2005).

Proposition 6.2 (Immediate possibility is not future geometry)**.** *Γ_t constrains local admissibility, whereas $\mathcal{S}_{h,\rho}(\mathcal{F}_t)$ represents recursively reachable structure over a non-zero horizon. The two coincide only in a grounding where the chosen horizon and representation reduce reachability to one local transition.*

7 Projection and recursion

The Seldon manifold may be viewed as a projection of recursively composed admissible transitions, but the future transition structure must be generated from each counterfactual successor rather than read as though the actual future were already known. Thus a schematic recursion is

$$\mathcal{S}_{h+1,\rho}(\mathcal{F}_t) = \mathcal{Q}_\rho\left(\bigcup_{\delta \in \mathcal{A}(\mathcal{F}_t)} \mathrm{Extend}\left(\delta, \mathcal{S}_{h,\rho}(\mathcal{F}_{t+1}^{(\delta)})\right)\right),$$

with boundary contingencies included in δ or in the successor frame as required by the grounding.

This is extensional. It does not require an implementation to enumerate every individual trajectory. Dynamic programming already exploits recursive state structure rather than recomputing complete histories independently (Bellman, 1957); set-based reachability methods similarly represent families of trajectories through shared geometric structure (Mitchell et al., 2005; Bansal et al., 2017).

8 Dynamic quantisation and temporal foveation

Hypothesis 8.1 (Adaptive resolution)**.** *For bounded representation, an effective future model assigns finer distinction where additional resolution materially changes reachable, acceptable, or controllable outcomes, and coarser distinction where refinement would not alter any operative decision.*

Thus $\rho(x, h, D)$ may increase where unresolved distinctions become decision-relevant. Near and distant horizons may use different quantisation, as may relevant and irrelevant spatial, causal, or conceptual regions. New dimensions may also be instantiated when a previously

absent distinction becomes operative; this is not a change from value zero but a change in representational dimension.

Adaptive mesh refinement provides an established computational analogue: resolution is recursively concentrated where local error estimates justify the additional cost (Berger and Oliger, 1984). The Seldon hypothesis differs in criterion—decision relevance rather than numerical truncation error—but shares the principle that uniform resolution is not mandatory.

9 Unresolved structure, and the present as a limit

Graphic Equalisation distinguishes four states of a dimension in a frame: NULL (not instantiated), undetermined (instantiated, no value realised), undisclosed (realised, not held by this frame), and resolved. That distinction is what the horizon-dependence of a Seldon manifold is actually about, and *which* of the two middle states is at issue determines which object one is talking about.

Definition 9.1 (Undetermined count)**.** *For frame $\mathcal{F}$ and projected horizon h, let*

$$U_{\mathcal{F}}(h)$$

be the number, or more generally the structure, of instantiated undetermined *dimensions in the frame's projection at that horizon — dimensions for which no value has been realised. NULL dimensions are excluded, being no variables at all;* undisclosed *dimensions are also excluded, being already determined.*

Remark (A manifold and a blind spot are different deficits)*.* A Seldon manifold is a projection over the *undetermined*: it represents continuations that have not been realised, and its width is the structure of what collapse has yet to fix. A blind spot (Definition 9.4) is a limitation over the *undisclosed*: the value exists and this frame does not hold it.

The two are frequently conflated under "uncertainty" and behave differently under every operation that matters. Undisclosed structure can be obtained — by observation, by another frame's report, or by inference from surrounding relations. Undetermined structure cannot be obtained by any of those, because there is nothing yet to obtain; it is settled only by collapse. A frame that improves its access reduces the second and leaves the first untouched.

Proposition 9.2 (The present is $U_{\mathcal{F}} = 0$)**.** *The present is the collapse boundary at which every instantiated undetermined dimension relevant to that collapse has been determined,*

$$\boxed{U_{\mathcal{F}} = 0,}$$

and as the projected horizon approaches the present,

$$U_{\mathcal{F}}(h) \longrightarrow 0.$$

Future distance in a Seldon manifold is therefore not merely geometric resolution. The operative quantity is *undetermined* structure — not what the frame has failed to observe, but what has not yet been settled.

Definition 9.3 (Observer-relative determinability)**.** *Let $D_{\mathcal{F}}(h)$ be the fraction, or measure, of determining structure resolved in-frame at horizon h, with*

$$\lim_{h \to 0^+} D_{\mathcal{F}}(h) = 1, \qquad D_{\mathcal{F}}(0) = 1.$$

Remark (What is changing, and what is not)*.* The Graph Law is deterministic throughout, at every horizon. What varies with h is how much determining structure is *resolved in-frame*. It is therefore incorrect to say that the universe becomes more deterministic as the present is approached; determinism is not a quantity that varies. The observer's determinability does.

Consistently with Graphic Equalisation's distinction between *determinable* and *determined*, a continuation may be fully determinable from present structure without having been realised. The manifold represents determinable continuations; collapse produces determined ones.

Definition 9.4 (Blind spot)**.** *A frame may fail to observe a projected region directly while still constraining its continuation through relations around it. Three states must therefore be distinguished:*

$$\boxed{\textit{directly resolved} \neq \textit{unresolved} \neq \textit{occluded but inferable.}}$$

A blind spot is a limitation of access, not a hole in reality, and an occluded-but-inferable region is not epistemically equivalent to an unapproached one. Collapsing the two discards the surrounding constraint.

All three states in Definition 9.4 concern the *undisclosed*: in each case a value has been realised and the question is only how much of it this frame can reach. None of them is a statement about the undetermined, and a projection that mixes the two will report a manifold width that no amount of observation could ever reduce, or an occlusion that no observation could ever resolve.

10 Reachability rather than point prediction

Point prediction seeks

$$\widehat{\mathcal{F}}_{t+h} \approx \mathcal{F}_{t+h}.$$

The Seldon formulation instead asks which futures remain reachable under the represented counterfactual family:

$$\mathcal{F}_{t+h} \in \mathcal{S}_{h,\rho}(\mathcal{F}_t).$$

This emphasis is close to viability and Hamilton–Jacobi reachability, where the central objects are sets of states that can remain safe, reach targets, or avoid undesirable regions under admissible inputs (Aubin, 1991; Mitchell et al., 2005; Bansal et al., 2017). A point forecast can still be useful, but it is a statistic or selected continuation over the reachable structure rather than the primary object.

11 Acceptable futures

Goals are optional represented capabilities.

Definition 11.1 (Acceptable manifold)**.** *Where a frame represents goals, viability conditions, risk bounds, or constraints,*

$$\boxed{\mathcal{G}_{h,\rho}(\mathcal{F}_t) \subseteq \mathcal{S}_{h,\rho}(\mathcal{F}_t)}$$

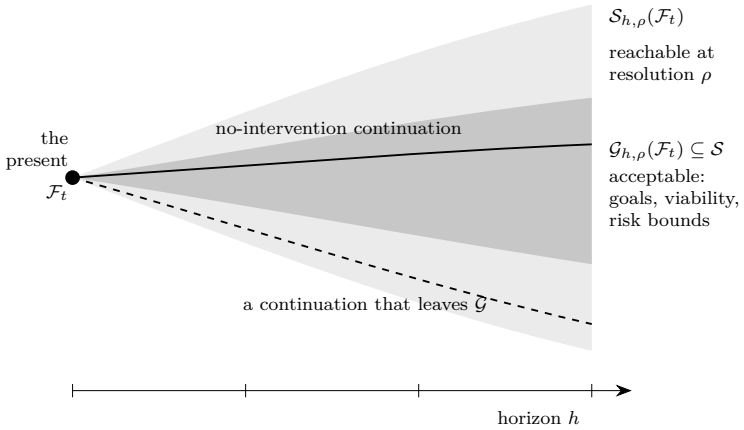
is the subset of reachable futures satisfying them.

This overlaps directly with viability kernels and feasible regions in constrained control and model-predictive control (Aubin, 1991; Rawlings et al., 2017; Cunis and Kolmanovsky, 2021). The Seldon terminology is broader because the acceptable subset may be defined over any represented future distinctions, not only a conventional continuous state space.

Figure 1 draws the construction.

Proposition 11.2 (Minimum intervention in a safe interior)**.** *Suppose the represented no-intervention alternative exists and its continuation remains within the acceptable manifold over the decision horizon. If every non-trivial intervention has non-negative cost and no independent objective rewards intervention itself, then a positive-cost intervention cannot dominate the no-intervention alternative solely by preserving acceptability.*

Remark. "No intervention" here is an instantiated alternative with a defined consequence, not NULL (Definition 5.1); the proposition does not identify absence with a zero-valued action.



Minimum intervention. If the no-intervention continuation stays inside $\mathcal{G}$ over the decision horizon, and every non-trivial intervention has non-negative cost, then intervening cannot dominate simply by preserving acceptability.

OBSERVER MACHINERY. This is a representation a frame may construct, not a structure the world contains. The temporal ratchet runs without it; nature does not have to compute a reachable set in order to evolve. Horizon distance here is unresolved structure ($U_{\mathcal{F}}(h) \to 0$ as $h \to 0$), not geometric distance — and the law is deterministic at every h.

THEORY FIGURE. The reachable set drawn here is not instrumented in the reference substrate: one call site audits ONE candidate trajectory, and the admissible-move enumeration is never composed with a deformation metric. Until they are joined, this geometry is a claim about what such a composition would compute — not a picture of a measurement.

Figure 1: The Seldon manifold, its acceptable subset, and the minimum-intervention proposition. The reachable set is a representation a frame may construct (Remark 4), not a structure the world contains.

12 Goal direction

Where local direction is meaningful, let $\mathbf{g}_t$ be a represented preferred direction through the acceptable region. A grounding may write

$$\mathbf{v}_{\mathcal{S}} \cdot \mathbf{g}_t > 0$$

for progress. No differentiable structure is required by the substrate; a graph grounding may instead use an ordering or directed edge relation.

13 Cost contours and basins

The manifold as defined says which continuations are reachable. It does not say that they are equally easy, and they are not. A manifold therefore admits a scalar field over it — a cost assigned to each represented continuation — and the geometry of that field is as informative as the extent of the manifold itself. What the scalar *is* is supplied by the companion papers, where surviving history-bearing structure prices continuations rather than removing them, and where the capacity to pay is itself a frame-relative quantity; the present paper needs only that some such field exists over $\mathcal{S}_{h,\rho}(\mathcal{F}_t)$ and is frame-relative in the same way the manifold is.

Definition 13.1 (Cost field and contours)**.** *Let $C_{\mathcal{F}}$ assign to each continuation in $\mathcal{S}_{h,\rho}(\mathcal{F}_t)$ the cost, in that frame's terms, of reaching it. Its level sets are the* contours *of the manifold, and the region enclosed by a contour is the set of continuations reachable within that budget. A* basin *is a region whose cheapest continuations lead back into it, so that leaving requires paying more than remaining.*

Two properties make this more than a way of drawing the manifold. The first is that the field has a default direction of travel. Retaining structure is free and altering it is priced, so a frame that does nothing does not stay still: it follows the descent, and the descent is where the narrative goes. Contours therefore make a projection visible that the bare reachable set does not

— not which futures are possible, which is the manifold's extent, but which are *cheap*, which is its shape.

The second is that basins are where the substrate's persistence phenomena live. A frame inside a basin will return to it under every cheap continuation, and will leave only by paying the ridge; the structural lock-in of the companion account is a basin read from above, and the horizon is the contour beyond which the required payment exceeds what the frame has. Whether a basin holds a given frame is therefore not a property of the basin alone but of the basin against that frame's capacity, which is why the same landscape confines one frame and not another.

Three consequences worth stating. Deformation, in the sense of the control section above, is re-shaping this field rather than moving within it: an intervention that changes what things cost changes where the descent leads, which is a larger change than moving a frame to a different point at fixed cost. Competing deformation is competing re-shaping, and the ridges are where the competition is decided. And a frame with enough capacity is not confined by any basin the landscape contains — basins say where the cheap path runs, never where a frame must go.

A cost field alone is not the whole of it, and taking it for the whole is the error the construction most invites. Cost is a flow — what reaching a continuation takes — while what matters equally is the state left behind it: the capacity a frame expects to have *after* arriving, in the terms the companion papers give that capacity. A full manifold carries both.

The two fields would collapse into one if capacity were purely depletive, so that what remains were simply what was held less what was spent. The companion account denies exactly this: an expenditure can net-gain, and capacity is not a stock drawn down. So the fields are independent, and their divergence carries the structure. An expensive continuation that raises capacity and a cheap one that lowers it are the two cases a single field cannot represent, and both are ordinary.

Definition 13.2 (Trap)**.** *A* trap *is a region in which the two fields anti-correlate: cheap to enter, capacity-reducing while occupied, and expensive to leave. A basin is merely a place the cheap path returns to; a trap is a basin whose occupancy erodes the means of leaving it.*

Two consequences follow, and they are the practical content of the section. Descent on cost alone is the wrong policy on a two-field landscape: it is greedy, and a frame that can look further will pay a ridge to reach a position of higher capacity, which is a move the cost field alone represents as simply worse. And the reversibility preference stated above acquires its reason here — a reversible action is one that preserves capacity, so preferring reversibility is preferring the second field where the first is indifferent, which is why the preference is not merely caution.

Honest boundary: this is a claim about the *shape* of the fields and not their values. Where the basins lie relative to a frame's capacity, where the two fields diverge, which ridges separate them, and which way the descent runs are the content; the metric that would put numbers on the contours is supplied by a grounding, if it is available at all, and nothing here asserts that the field is computable for a given system. Nor does the field add anything to the manifold's extent: a continuation priced beyond any available capacity remains reachable in principle and is excluded by cost, not by admissibility, which is the distinction the companion law turns on.

14 Muchness

Definition 14.1 (Muchness)**.** *Muchness*

$$\boxed{\mathcal{M}(\mathcal{F}_t; h, \rho) \in \mathbb{Z}}$$

is a frame-relative discrete measure of materially distinguishable future possibility retained within $\mathcal{S}_{h,\rho}(\mathcal{F}_t)$*.*

A grounding supplies a discrete measure

$$\mu_{\mathcal{M}} : \mathcal{Q}_\rho(\mathcal{S}) \to \mathbb{Z}$$

that preserves the distinctions relevant to the claimed domain. It may count distinguishable states, equivalence classes, connected reachable regions, retained capability classes, or use another integer index. Muchness therefore measures *how much distinguishable future remains*, not how strongly the frame can alter that future.

Proposition 14.2 (Resolution and dimension dependence)**.** *Muchness is not meaningful without the frame, represented dimensions, horizon, and resolution that determine which future distinctions exist and which count as distinct.*

The term *Muchness* and its role are specific to this framework. Classical reachable-set and viability theory provide the geometry from which such a measure can be constructed but do not identify this quantity under that name.

15 Irreversibility

Definition 15.1 (Irreversible contraction)**.** *An intervention is irreversible relative to frame, horizon, dimensions, and resolution when it makes previously distinguishable valuable future regions unreachable and no admissible continuation within the stated horizon restores them.*

Where the Muchness dimension is instantiated before and after the intervention, a simple discrete contraction measure is

$$I_{\mathcal{F}}(u) = \max\Big(0, \mathcal{M}_t - \mathcal{M}_{t+1}^{(u)}\Big)\,.$$

Here 0 is a legitimate value on an existing irreversibility dimension, not absence of the dimension (Definition 5.1).

The option value of preserving reversible choices under uncertain future information has a long history in economics (Arrow and Fisher, 1974; Henry, 1974; Dixit and Pindyck, 1994). The present definition is more primitive: it identifies irreversibility structurally with loss of reachable distinctions. No substrate-level probability distribution is required. If a frame overlays probabilities on unresolved future conditions, an expected irreversibility statistic is an epistemic derived quantity, not part of the deterministic Graph Law.

Proposition 15.2 (Reversibility preference)**.** *If two interventions achieve the same represented objective to equivalent tolerance and otherwise differ only in that one preserves a strict superset of potentially valuable future distinctions, any represented valuation assigning non-negative value to retained optionality weakly prefers the more reversible intervention.*

16 Control as manifold deformation

Definition 16.1 (Control)**.** *Where intervention is represented, control is deformation of reachable future geometry:*

$$u : \mathcal{S} \mapsto \mathcal{S}'.$$

A controller need not predict one exact future trajectory. It may succeed by deforming reachable geometry so that realised collapse remains inside an acceptable region while preserving useful future distinctions. This set-oriented view is close to viability regulation, reachability-based safety control, and constrained model-predictive control (Aubin, 1991; Mitchell et al., 2005; Rawlings et al., 2017; Cunis and Kolmanovsky, 2021).

17 Competing deformation

Several processes may shape one shared Seldon manifold. A local additive approximation

$$\Delta\mathcal{S} \approx \sum_i \Delta\mathcal{S}_i$$

may be useful where interactions are approximately separable. More generally,

$$\Delta\mathcal{S} = \Phi(u_1, \ldots, u_n; G, \Gamma, W).$$

Dynamic-game reachability provides established mathematics for reachable sets under interacting or adversarial controls (Mitchell et al., 2005). The Seldon statement is more general and does not require zero-sum or game-theoretic valuation.

This shared-arena case differs from one graph individuated by several frames: each individuating frame carries its own represented dimensions, resolution, and reachable geometry unless a containing relation explicitly couples them.

18 Reachability compression

Let $\{\gamma_i\}$ be an explicit family of trajectories.

Hypothesis 18.1 (Reachability compression)**.** *For task classes in which many trajectories are equivalent with respect to the operative distinctions at resolution ρ, a representation of their shared reachable geometry can have lower decision-relevant description complexity than explicit trajectory enumeration:*

$$L(\mathcal{S}) \ll L(\{\gamma_i\}).$$

This hypothesis is compatible with the general minimum-description-length principle that useful models exploit shared regularity rather than restating each observation independently (Rissanen, 1978). It is also reflected constructively in set-based reachability methods, which compute or approximate reachable regions without storing every individual trajectory (Mitchell et al., 2005; Maidens et al., 2013). The stronger claim that Seldon representations systematically achieve lower *decision-relevant* description cost is specific to this paper and remains falsifiable.

19 Relationship to Narrativium and Narrative

The present paper distinguishes retained distinguishable future possibility ($\mathcal{M}$) from the capacity to deform it (N); the contrast is set out in full in the companion Narrativium paper, which develops N. Narrative is also unnecessary for defining $\mathcal{S}$, but where a containing frame carries operative history-bearing state, that state may orient the manifold; this is the subject of Narrative Momentum.

20 Causal depth is not temporal distance

The ordinary picture of time,

$$\text{history} \rightarrow \text{now} \rightarrow \text{future},$$

is misleading as a picture of what determines a continuation, and the correction is structural rather than presentational.

Determination has two components. One is *succession*: the order realised changes carry through their enabling-relations, which is what makes a time dimension meaningful at all. The other is *structural depth*: relations instantiated in containing frames, which are present now and are not earlier in time.

Figure 2 contrasts the two pictures directly.

Remark (The tilt). Replacing time by structure entirely would be a 90° rotation of the causal picture and is not what is claimed. The correct adjustment is closer to a 45° change of causal vector: succession still runs, and additional determining structure is reached by depth rather than by reaching further back.

Remark (No dependent clocks: a subframe need not be slaved to its container). Because depth and succession are orthogonal, structural containment carries no temporal subordination. A frame's time is its own change-count (Graphic Equalisation, Axiom on time), and with no global clock there is nothing for a contained frame's clock to be slaved to; each frame counts its own changes. The intuition that a contained process runs *within* its container's time is the assumption of a dependent clock, and it is a property of embodiments that happen to share one, not a substrate fact. Consequently a subframe can run *faster* than its superframe — accumulate more proper time over the same stretch — when more of the fixed change-budget falls in it. This does not exceed c: the change-rate c is the same in every frame (Graphic Equalisation, c as the change-to-time rate), and what is free is the *ratio* of change-counts between frames, the distribution of the budget across regions, not the per-change rate. A gravitational subframe's ratio is bounded by its physics and a computational subframe's by its throughput; the substrate imposes no dependent-clock law, and the achievable ratio is an engineering fact of the embodiment.

The complement holds equally, and completes the picture: dependence is not *forbidden* either. A subframe *may* be driven — ticked by a containing frame, and by one several levels above it — so its clock is then slaved to that superframe's; this is the coupled, locally synchronous case, and it is a coupling paid for, not a substrate necessity. Clock-coupling between a subframe and a container is therefore a free parameter running from fully independent (asynchronous) to fully ticked (synchronous), and nothing compulsory fixes where a given subframe sits. Reality samples the whole range, which is Graphic Equalisation's globally-asynchronous, locally-synchronous structure read on the clock: bound and driven subframes are ticked; uncoupled ones run their own change.

Proposition 20.1 (Depth comes from superframes, not from historical frames)**.** *Where a continuation is determinable beyond what $\mathcal{F}_0$ resolves, the additional determining structure is supplied by containing frames*

$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots ,$$

potentially without a privileged terminal frame. It is not *supplied by retained temporal snapshots: by Graphic Equalisation's history principle, past states influence the present only where they survive as present structure.*

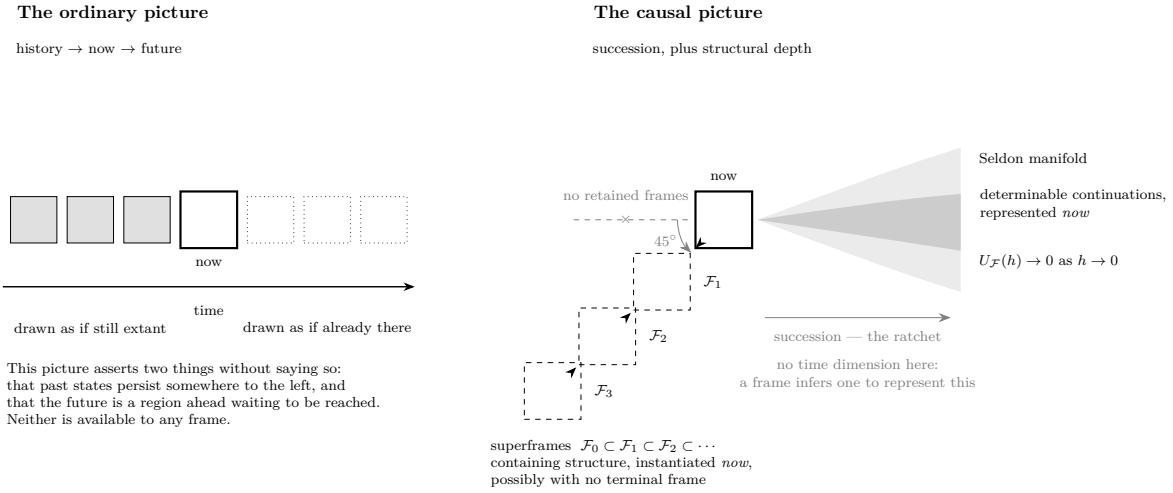
This matters for what an explanation is permitted to assume. A theory that located determining structure in earlier temporal states would require arbitrarily deep access to a past that no frame holds. Locating it in containing frames requires only structure that is instantiated now.

21 Positioning and novelty

The paper's relationship to established work can be summarised as follows.

Established prior art.

Reachable sets, viability kernels, invariant and capture sets, dynamic programming, model-predictive feasible sets, Hamilton–Jacobi reachability, adaptive numerical resolution, and option value under irreversibility are established (Bellman, 1957; Aubin, 1991; Mitchell et al., 2005; Berger and Oliger, 1984; Arrow and Fisher, 1974; Rawlings et al., 2017).



The tilt. Replacing time by structure entirely would be a 90° rotation, and is not the claim. Succession still runs. What changes is where the *additional* determining structure is found — in containing frames, instantiated now, rather than in temporal states that no longer exist. Roughly a 45° change of causal vector, not a 90° replacement of time by structure.

On the axis. The left panel's "time" is the ordinary picture's own assumption. The right panel has succession and no time dimension: succession is a relation between realised collapses, and a *time dimension* is what an inferring frame constructs to represent it.

THEORY FIGURE. The manifold is observer machinery — a representation a frame may construct, not a structure the world contains. The ratchet runs without it. Nothing in this diagram is instrumented in the reference substrate.

Figure 2: The ordinary picture of time against the causal picture. Left: past states drawn as though still extant and the future drawn as a region already there — neither of which any frame holds. Right: succession forward, structural depth through containing frames at roughly 45°, and the manifold as observer machinery ahead of the present.

Structural synthesis.
The Seldon manifold treats these set-valued future structures as a frame-relative representation generated downstream of deterministic Graphic Equalisation and evaluated only on instantiated dimensions at operative discrete resolution.

Specific contributions.
The terminology and formal roles of the Seldon manifold and Muchness; the explicit separation of local admissibility, counterfactual reachable geometry, deterministic collapse, Muchness, and Narrativium; dimension-instantiation discipline (NULL is not zero); and the decision-relevant reachability-compression hypothesis are claims of this framework.

22 Falsifiability commitments

1. **Cost contours and basins.** Fails if the continuations a frame actually takes, absent intervention, are not the cheap ones — if the descent direction of the cost field does not predict where an uninterrupted frame goes — or if persistence in a region can be exhibited with no corresponding basin, or a basin with no corresponding persistence. It fails also if the two fields cannot come apart — if expected capacity after an action is always the capacity before it less the cost, so that no expensive continuation ever raises capacity and no cheap one lowers it. (The claim is about the fields' shape relative to a frame's capacity, not about the values of any metric over them.)

2. **Deterministic counterfactual typing.** Fails if the Seldon construction requires multiple fundamentally random realised outcomes from one causally complete antecedent rather than multiple deterministic continuations under differing represented future conditions.

3. **Reachability utility.** Fails for a claimed task class if reachable geometry cannot preserve

distinctions needed for prediction or control that trajectory-level representation preserves.

4. **Dimension/value typing.** Fails if an absent future dimension must be represented as the numerical value zero rather than remaining undefined until instantiated.

5. **Adaptive resolution.** Fails as an efficiency hypothesis if decision-directed refinement gives no advantage over uniform refinement in the claimed class.

6. **Muchness distinction.** Fails if retained distinguishable possibility and capacity to deform that possibility cannot be independently varied in any grounding.

7. **Irreversibility.** Fails for a claimed grounding if supposedly deleted reachable distinctions are always restorable without crossing an additional constraint or expending additional compatible capability.

8. **Reachability compression.** Fails if explicit trajectory representations consistently achieve equal or lower decision-relevant description cost for the claimed task families.

23 Conclusion

Graphic Equalisation supplies deterministic admissible continuation and collapse. The Seldon manifold is the structured geometry of unrealised counterfactual continuation that remains reachable:

$$\boxed{\Gamma \to \text{admissible possibility} \to \mathcal{S} \to \text{reachable geometry} \xrightarrow{\mathcal{C}} \text{realised history.}}$$

The distinction between reachable geometry and realised history is essential: the manifold contains counterfactual futures because the frame has not fixed all future interventions and containing conditions, not because the Graph Law is stochastic.

The future, in this framework, is not primarily a destination. It is a reachable geometry.

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